

HKCEE 1981

Mathematics II

81
1. $\frac{(a^2b^{-3})^2}{a^{-2}b} =$

- A. a^2b^{-7}
 B. a^2b^{-5}
 C. a^6b^{-2}
 D. a^6b^{-6}
 E. a^6b^{-7}

- A. $2^{(x^x)}$
 B. $2^x \cdot x^x$
 C. $2x^x$
 D. 2^{2x}
 E. $2^{(x^2)}$

81
2. $\frac{1}{x+1} + \frac{1}{x-1} + \frac{x+\frac{1}{x}}{x-\frac{1}{x}} =$

- A. $\frac{1}{x+1}$
 B. $\frac{1}{x-1}$
 C. $\frac{x+1}{x-1}$
 D. $\frac{1}{(x+1)(x-1)}$
 E. $\frac{x^2+4x+1}{(x+1)(x-1)}$

81
5. $\left(\frac{\frac{x}{y} + \frac{y}{x} + 2}{\frac{x}{y} - \frac{y}{x}} \right)^{-1} =$

- A. $\frac{x-y}{x+y}$
 B. $\frac{x+y}{x-y}$
 C. $-\frac{x+y}{x-y}$
 D. $\frac{x^2+y^2}{x^2-y^2}$
 E. $\frac{x^2-y^2}{x^2+y^2}$

81
3. If $x = \frac{-bx+ay-c}{a+by}$, then $y =$

- A. $\frac{ax+bx+c}{a-bx}$
 B. $-\frac{ax+bx+c}{a-bx}$
 C. $\frac{ax+bx+c}{a+bx}$
 D. $-\frac{ax+bx+c}{a+bx}$
 E. $\frac{ax-bx-c}{a-bx}$

81
6. If $H = K + \frac{M}{4\pi(r^2+l^2)^n}$ and $r > 0$, then
 $r =$

- A. $\left\{ \left[\frac{M}{4\pi(H-K)} \right]^{-n} - r^2 \right\}^{\frac{1}{2}}$
 B. $\left[\frac{M}{4\pi(H-K)} \right]^{\frac{n}{2}} - l$
 C. $\left\{ \left[\frac{M}{4\pi(H-K)} \right]^{\frac{1}{n}} - l^2 \right\}^{\frac{1}{2}}$

81
4. $(2^x)^x =$

D. $\left[\frac{M}{4\pi(H-K)} \right]^{\frac{1}{2n}} - l$

E. $\left\{ \left[\frac{4\pi}{M(H-K)} \right]^{\frac{1}{n}} - l^2 \right\}^{\frac{1}{2}}$

81 If $f(x) = x^2 + x + 1$, then $f(x+1) - f(x)$
7.

- A. 1
- B. 3
- C. $2x + 1$
- D. $2x + 2$
- E. $x^2 + x + 1$

81 If $\log_{10}x + \log_{10}4 = \log_{10}(x+4)$, what is
8. the value of x ?

- A. 0
- B. 1
- C. $\frac{4}{3}$
- D. 4
- E. x may be any positive number

81 It is given that
9. $x(2x+3) = x(3x-4)$. $x = ?$

- A. 0 only
- B. 7 only
- C. 0 or 7
- D. $-\frac{3}{2}$ or $\frac{4}{3}$ only
- E. $0, -\frac{3}{2}$ or $\frac{4}{3}$

81 $2y - 3 > 4y + 2x + 5$ is equivalent to
10.

- A. $y > x + 4$
- B. $y < x + 4$
- C. $y > -x - 4$
- D. $y < -x - 4$
- E. $y > x + 1$

81 The n th term of the arithmetic
11. progression 2, 6, 10, 14, ... is

- A. $2n^2$
- B. $4n$
- C. $4n - 2$
- D. $4n + 2$
- E. $6 - 4n$

81 If $3x - 2y = x + 3y$, then $x^2 : y^2 =$
12.

- A. 2 : 5
- B. 5 : 2
- C. 4 : 25
- D. 25 : 4
- E. 1 : 4

81 The marked price of a book is $\$x$. 30%
13. of this price is profit. If the book is sold at a discount of 20%, what will the profit then be?

- A. $\$0.04x$
- B. $\$0.06x$
- C. $\$0.1x$
- D. $\$0.24x$
- E. $\$0.56x$

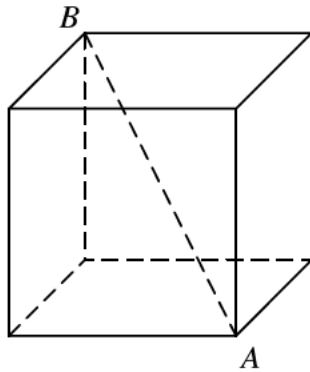
81 A group consists of n boys and n girls.
14. If two of the girls are exceeded by two other boys, then 51% of the group members will be boys. What is n ?

- A. 50
- B. 51
- C. 52
- D. 100
- E. 102

81 If the surface area of a spherical soap
15. bubble increases by 44%, its volume increases by

- A. 20%
- B. 33.1%
- C. 60%
- D. 66%
- E. 72.8%

81
16.



The total area of the six faces of the solid cube in the figure is 96 cm^2 . What is the length of the diagonal AB ?

- A. $6\sqrt{2} \text{ cm}$
- B. $4\sqrt{3} \text{ cm}$
- C. $4\sqrt{2} \text{ cm}$
- D. $2\sqrt{6} \text{ cm}$
- E. 4 cm

81
17. A merchant sold 100 chairs. 80 of them were sold at a profit of 30% on each chair, while 20 of them were sold at a loss of 40% on each chair. What is his percentage gain or loss on the whole stock?

- A. A loss of 8%
- B. A loss of 10%
- C. A gain of 8%
- D. A gain of 16 %
- E. A gain of 24%

81
18. If $0^\circ < \theta < 90^\circ$ and $\sin \theta = \frac{k}{2}$, then $\cos \theta =$

- A. $1 - \frac{k}{2}$
- B. $\frac{2}{\sqrt{4+k^2}}$
- C. $\frac{\sqrt{4+k^2}}{2}$

- D. $\frac{2}{\sqrt{4-k^2}}$
- E. $\frac{\sqrt{4-k^2}}{2}$

81
19. $\tan \theta \sin \theta - \frac{1}{\cos \theta} =$

- A. 0
- B. $\cos \theta$
- C. $-\cos \theta$
- D. $\frac{-1}{\cos \theta}$
- E. $-\tan \theta \sin \theta$

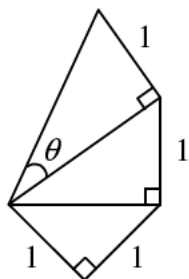
81
20. If $0^\circ \leq \theta \leq 360^\circ$, the number of roots of the equation $2 \sin \theta \cos \theta - \cos \theta = 0$ is

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

81
21. An angle measures x radians. What is its measure in degrees?

- A. $\left(\frac{\pi x}{180}\right)^\circ$
- B. $\left(\frac{180x}{\pi}\right)^\circ$
- C. $\left(\frac{\pi}{180x}\right)^\circ$
- D. $\left(\frac{\pi x}{360}\right)^\circ$
- E. $\left(\frac{360x}{\pi}\right)^\circ$

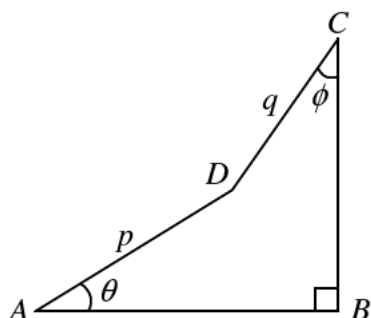
81
22.



In the figure, $\cos \theta =$

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. $\frac{3}{4}$
- D. $\frac{\sqrt{3}}{2}$
- E. $\frac{\sqrt{3}}{4}$

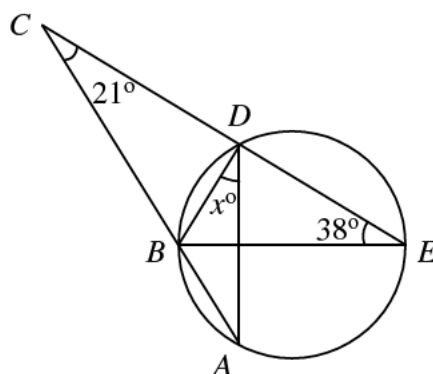
81
23.



In the figure, $AD = p$, $DC = q$, $\angle B = 90^\circ$. $AB =$

- A. $p \sin \theta + q \sin \phi$
- B. $p \cos \theta + q \cos \phi$
- C. $p \sin \theta + q \cos \phi$
- D. $p \cos \theta + q \sin \phi$
- E. $(p + q)(\cos \theta + \cos \phi)$

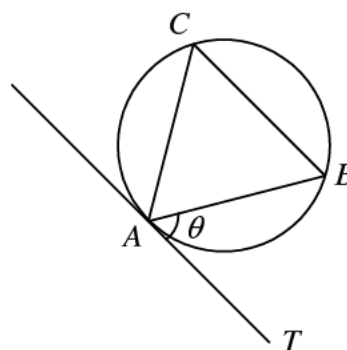
81
24.



In the figure, BE is a diameter of the circle. ABC and EDC are straight lines. $x^\circ =$

- A. 21°
- B. 31°
- C. 38°
- D. 52°
- E. 59°

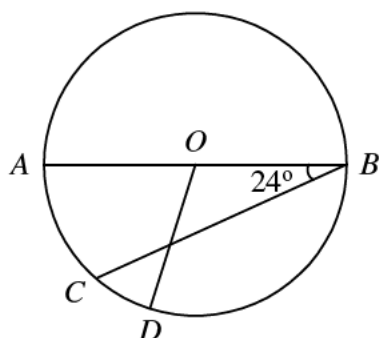
81
25.



In the figure, AT touches the circle at A . In $\triangle ABC$, $\angle A : \angle B : \angle C = 2 : 3 : 4$. $\theta =$

- A. 40°
- B. 50°
- C. 60°
- D. 70°
- E. 80°

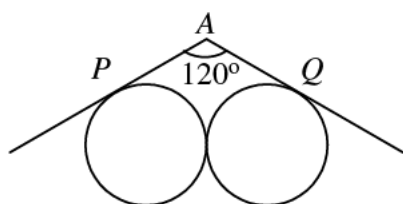
81
26.



In the figure, AB is a diameter of the circle with centre at O . The length of the minor arc AC is twice the length of the minor arc CD . $\angle BOD =$

- A. 72°
- B. 90°
- C. 108°
- D. 132°
- E. 144°

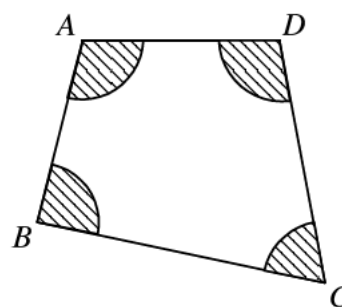
81
27.



In the figure, two circles both with radius 2 cm touch each other externally. AP and AQ are equal tangents to the two circles. $AP = ?$

- A. $\sqrt{3}$ cm
- B. $2\sqrt{3}$ cm
- C. 4 cm
- D. $4\sqrt{3}$ cm
- E. $\frac{4\sqrt{3}}{3}$ cm

81
28.



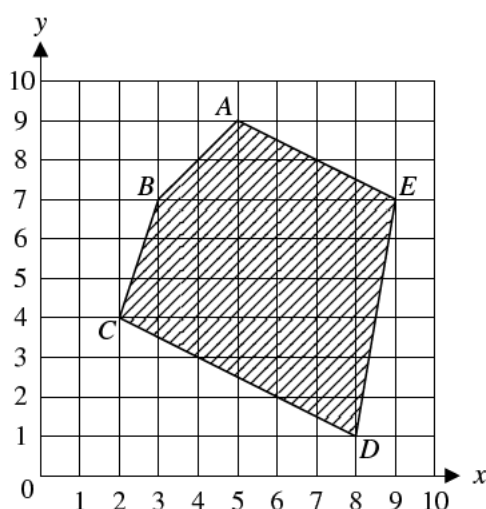
In the figure, $ABCD$ is a quadrilateral. The shaded portions are four sectors with centres at A, B, C and D . Their radii are all equal to a . What is the total area of the four sectors?

- A. πa^2
- B. $2\pi a^2$
- C. $4\pi a^2$
- D. $\sqrt{2} \pi a^2$
- E. It cannot be determined

81
29. $2x^2 - 2 \leq 0$ is equivalent to

- A. $x \leq 1$
- B. $x \geq -1$
- C. $-1 \leq x \leq 1$
- D. $x \geq 1$ or $x \leq -1$
- E. $x \leq 1$ or $x \geq -1$

81
30.



In the figure, which point in the shaded region will make the value of $x - 2y$ a minimum

- A. A
- B. B
- C. C
- D. D
- E. E

81 $6x^2 + kx + 6 = 0$ is a quadratic equation
31. in which k is a constant. Its roots α and β are positive. $\log_{10}\alpha + \log_{10}\beta =$

- A. 0
- B. 1
- C. $\log_{10}6$
- D. $\log_{10}(-k)$
- E. $\log_{10}\left(-\frac{k}{6}\right)$

81 $-3x^2 - 3x \equiv -3(x + a)^2 + b$ is an identity
32. in x . What are the values of the constants a and b ?

- A. $a = 1$ and $b = 0$
- B. $a = \frac{1}{2}$ and $b = \frac{3}{4}$
- C. $a = \frac{1}{2}$ and $b = \frac{3}{4}$
- D. $a = -\frac{1}{2}$ and $b = \frac{3}{4}$
- E. $a = -\frac{1}{2}$ and $b = -\frac{3}{4}$

81 The H.C.F. and L.C.M. of three
33. expressions are a^2b^2c and $a^4b^6c^4$ respectively. Two of the expressions are $a^2b^3c^4$ and $a^3b^2c^2$. The third expression is

- A. a^3b^3c
- B. $a^3b^6c^4$
- C. a^4b^2c
- D. a^4b^6c
- E. $a^4b^6c^2$

81 The sum of the first five terms of an
34. arithmetic progression is 15. If the fourth term is 7, the first term is

- A. -5
- B. -3
- C. -1
- D. 1
- E. 10

81 Which of the following can be summed
35. to infinity?

- I. The arithmetic progression
4, 3, 2, 1,
- II. The geometric progression
27, 9, 3, 1,
- III. The geometric progression
16, -8, 4, -2,

- A. II only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

81 The running speeds of three boys A, B
36. and C are in the ratios $a : b : c$. The times that A, B and C take to complete a 1500 m race are in the ratios

- A. $a : b : c$
- B. $c : b : a$
- C. $b + c : a + c : a + b$
- D. $\frac{1}{a} : \frac{1}{b} : \frac{1}{c}$
- E. $\frac{a}{b} : \frac{b}{c} : \frac{c}{a}$

81 If n is a positive integer, which of the
37. following numbers is/are odd?

- I. 2^{2n+1}
- II. $3(2^n)$
- III. $(2n+1)^2$

- A. II only
- B. III only
- C. I and III only
- D. II and III only
- E. I, II and III

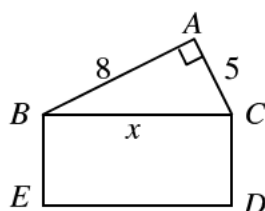
- 81 A factory employs x workers each
38. working n hours a day. The whole
factory produces k watches per day. If
 y workers go on leave, then how many
hours a day should the remaining
workers work in order to produce the
same number of watches per day?

- A. $\frac{nx}{y}$
B. $\frac{ny}{x}$
C. $\frac{nx}{4y}$
D. $\frac{nx}{x-y}$
E. $\frac{n(x-y)}{x}$

- 81 The daily wages of a man and a boy are
39. in the ratio 2 : 1. In a day a man has to
work 8 hours but a boy only 6 hours.
The hourly wages of a man and a boy
are in the ratio

- A. 8 : 3
B. 2 : 1
C. 3 : 2
D. 4 : 3
E. 1 : 1

81
40.

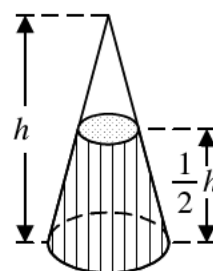


In the figure, $\angle BAC = 90^\circ$, $AB = 8$,
 $AC = 5$ and $AX \perp BC$. $BCDE$ is a
rectangle with $CD = AX$. What is the
area of the rectangle $BCDE$?

- A. 20
B. 40
C. 80
D. 89

E. $4\sqrt{89}$

81
41.

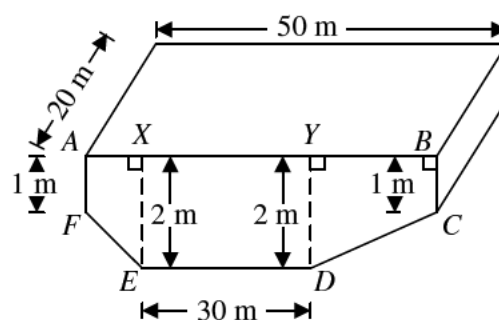


The height of the cone in the figure is
 h . It contains water to a depth of $\frac{1}{2}h$.

$$\frac{\text{Volume of water}}{\text{Capacity of the cone}} =$$

- A. $\frac{1}{8}$
B. $\frac{1}{4}$
C. $\frac{1}{2}$
D. $\frac{3}{4}$
E. $\frac{7}{8}$

81
42.



The figure above represents a $50\text{m} \times 20\text{m}$ swimming pool. The pool is in the shape of a prism with a rectangular surface and four vertical walls. The dimensions of the sidewall $ABCD$ are as shown in the figure. What is the capacity of the pool in m^3 ?

- A. 1200
B. 1500

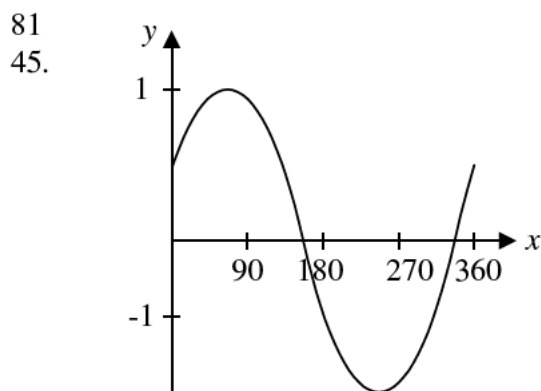
- C. 1800
- D. 2000
- E. It cannot be determined

81
43. Given that $\sin \theta - \cos \theta = \frac{1}{2}$, what is the value of $\sin \theta \cos \theta$?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{3}{8}$
- D. $\frac{3}{4}$
- E. It cannot be determined

81
44. If $0^\circ \leq \theta \leq 360^\circ$, the minimum value of $1 + 2\cos \frac{\theta}{2}$ is

- A. -2
- B. -1
- C. 0
- D. 1
- E. 2



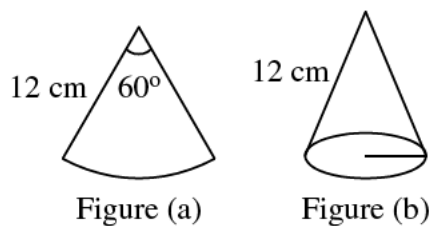
The figure above shows the graph of

- A. $y = \sin(x^\circ + 30^\circ)$
- B. $y = \sin(x^\circ - 30^\circ)$
- C. $y = \sin(x^\circ + 150^\circ)$
- D. $y = \sin(x^\circ - 150^\circ)$
- E. $y = \sin(x^\circ + 60^\circ)$

81
46. The radius of a sector is 3 cm and the perimeter is 10 cm. What is the area of the sector?

- A. 6 cm^2
- B. 12 cm^2
- C. 15 cm^2
- D. 18 cm^2
- E. 45 cm^2

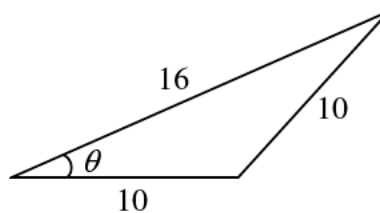
81
47.



The cone in Figure (b) is formed by bending the sector in Figure (a). The angle of the sector is 60° and the radius is 12 cm. The radius of the base of the cone is

- A. 2 cm
- B. 4 cm
- C. 6 cm
- D. $2\pi \text{ cm}$
- E. $\frac{360}{\pi} \text{ cm}$

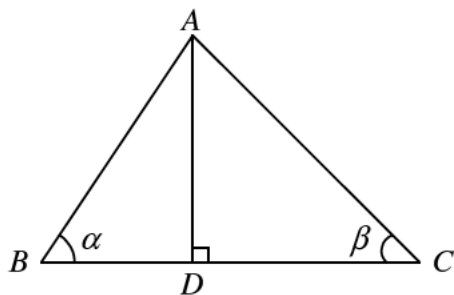
81
48.



In the figure, $\sin \theta =$

- A. 0.5
- B. 0.6
- C. 0.625
- D. 0.75
- E. 0.8

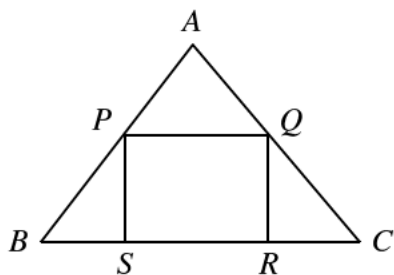
81
49.



In the figure, $AD \perp BC$. $CD =$

- A. $h \sin \alpha \tan \beta$
- B. $h \cos \alpha \tan \beta$
- C. $h \tan \alpha \sin \beta$
- D. $\frac{h \cos \alpha}{\tan \beta}$
- E. $\frac{h \sin \alpha}{\tan \beta}$

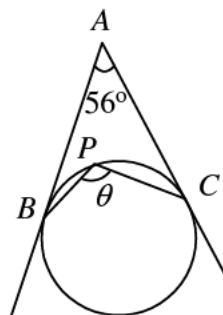
81
50.



In the figure, ABC is an equilateral triangle of side $2a$. P and Q are the mid-points of AB and AC respectively. $PQRS$ is a rectangle. What is the area of $PQRS$?

- A. a^2
- B. $\frac{1}{2}a^2$
- C. $\frac{2}{3}a^2$
- D. $\frac{1}{\sqrt{3}}a^2$
- E. $\frac{\sqrt{3}}{2}a^2$

81
51.

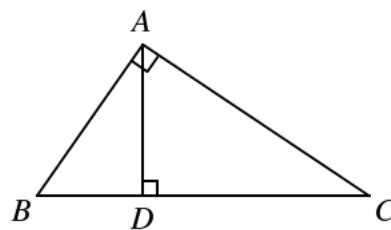


In the figure, AB and AC touch the circle at B and C . If P is any point on the minor arc BC , what is θ ?

- A. 112°
- B. 118°
- C. 124°
- D. 146°
- E. It cannot be determined

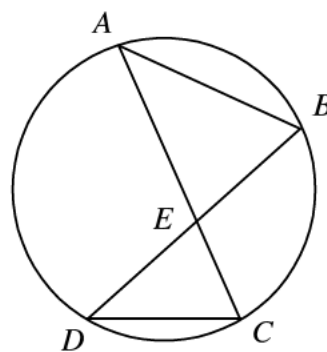
81
52.

I



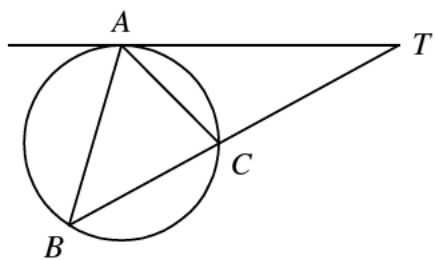
$\angle BAC = 90^\circ$
 $AD \perp BC$.

II



AC and BD
intersect at E .

III

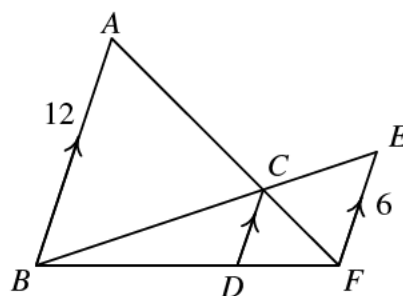


BC produced meets the tangent AT at T .

Which of the above figures contains one or more pairs of similar triangles?

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

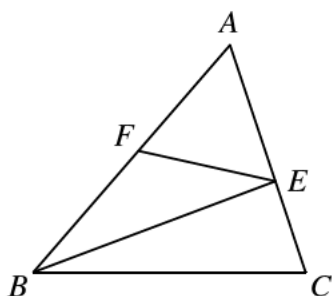
81
54.



In the figure, $AB \parallel CD \parallel EF$. ACF , BCE and BDF are straight lines.
 $AB = 12$, $EF = 6$. $CD = ?$

- A. 4.5
- B. 4
- C. 3.6
- D. 3
- E. 2

81
53.



In the figure, P is the mid-point of AB .
 E is a point on AC such that

$$AE : EC = 2 : 1. \quad \frac{\text{Area of } \triangle BFE}{\text{Area of } \triangle BCE} =$$

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. 1
- D. $\frac{3}{2}$
- E. 2